

Mapping with known Poses

Occupancy Grid Mapping

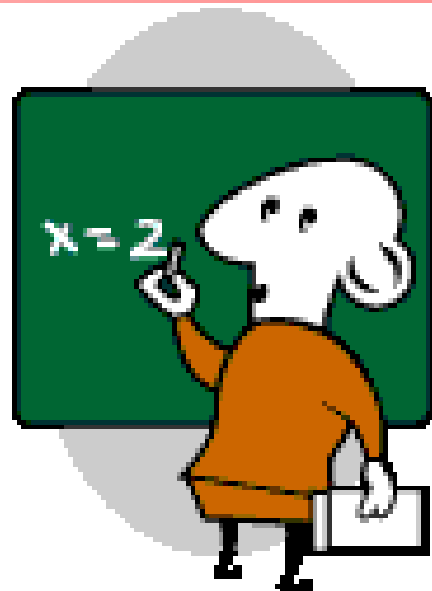
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*Lecture slides heavily use material from the textbook and
Sebastian Thrun, Lecture Slides; <http://www.probabilistic-robotics.org/>*



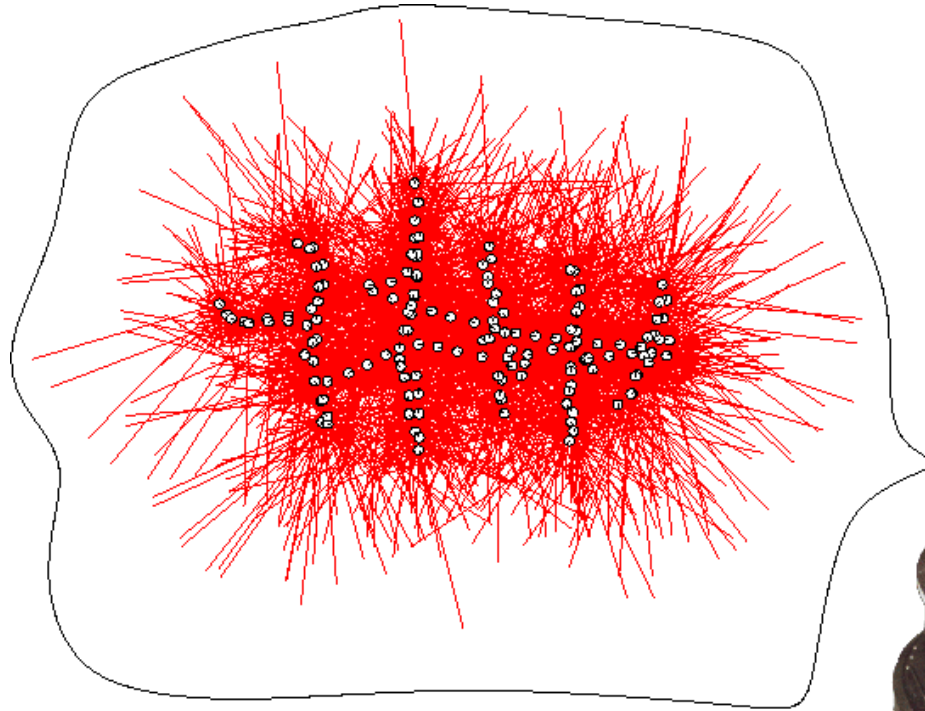
What we will discuss

- Define the mapping problem,
- Examine mapping with a “localized” robot,
- Occupancy Grid Mapping,
- Examples
- Multi-Sensor Fusion for Mapping,
- Learning Inverse Measurement Models,
- Maximum A posteriori (MAP) Occupancy Mapping





The Mapping Problem



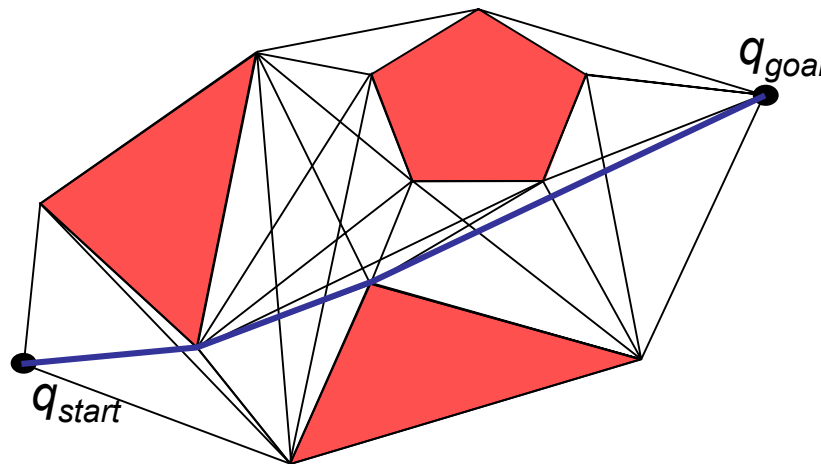
What does the environment look like?





Why Mapping?

- Learning maps is one of the fundamental problems in mobile robotics
- Maps allow robots to efficiently carry out their tasks, allow continuous localization ...
- Successful robot systems rely on maps for localization, path planning, activity planning etc.





The Mapping Problem

- Formally, mapping involves, given the sensor and actuator data,

$$d = \{u_1, z_1, u_2, z_2, \dots, u_n, z_n\}$$

- task of calculating the most likely map:

$$m^* = \arg \max_m P(m \mid d)$$

- For now, we will assume we are rather given

$$d = \{x_1, z_1, x_2, z_2, \dots, x_n, z_n\}$$



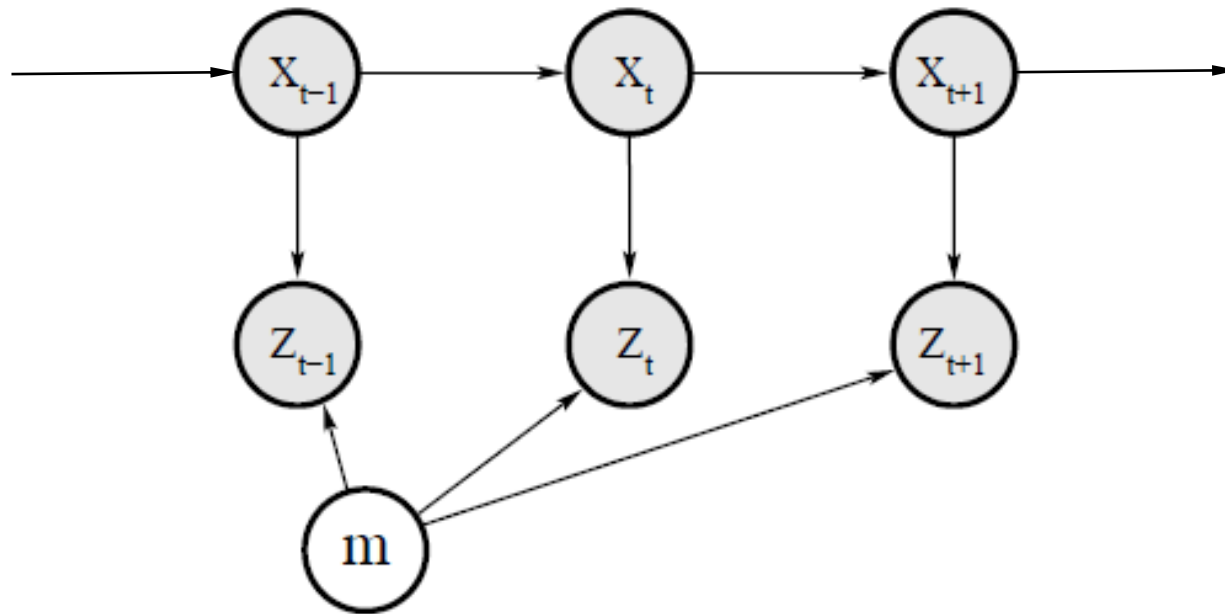
The Mapping Problem

- So far we learned how to estimate the pose of the vehicle given the data d and the map.
- True mapping, however, involves simultaneously estimating the pose of the vehicle and the map.
- The general problem is therefore denoted as the ***SLAM - simultaneous localization and mapping problem***
- Throughout this section we will describe how to calculate a map given we know the pose of the vehicle.



The Mapping Problem

- Dynamic Bayes Network graphical model of mapping





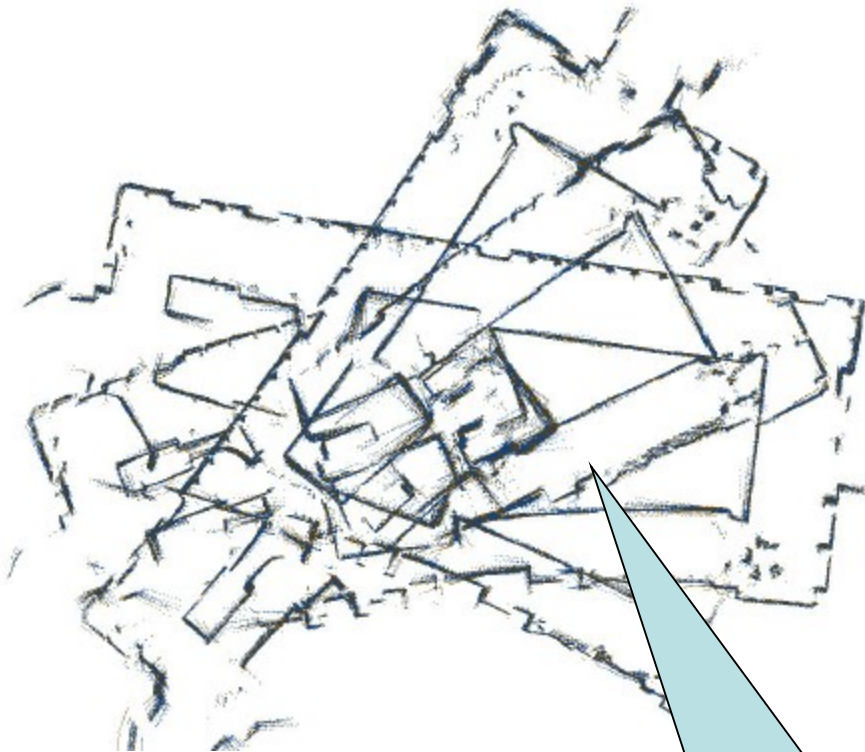
Nature of Mapping Problems

- **Size in relation to perceptual range,**
(The larger the environment, the more difficult the problem)
- **Noise in perception and actuation,**
(The larger the noise, the more difficult the problem)
- **Perceptual ambiguity**
(The more similarity/symmetry in the environment, the more difficult to find correspondences, hence the problem)
- **Cycles**
(Cycles are especially problematic because of the delay and difficulty in finding corresponding places)

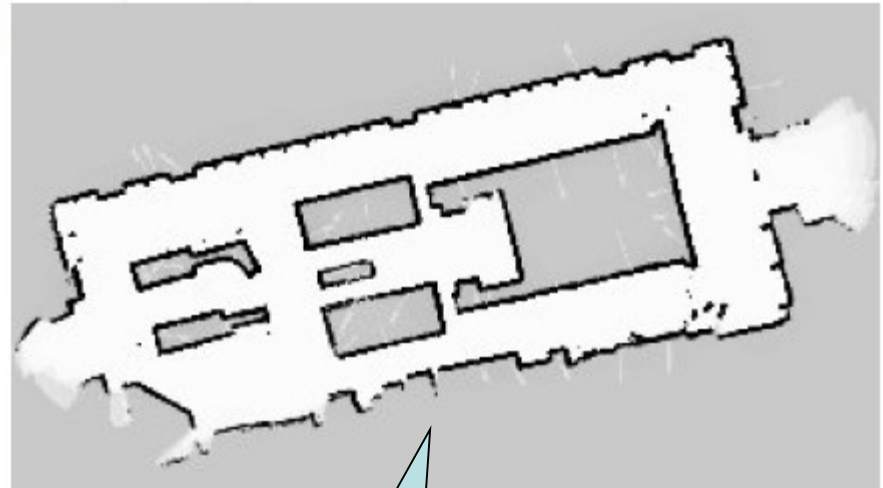


Example

(a)



(b)



Measurements indexed
(and merged) by odometry

Occupancy Grid Map



Sub-Problems of Mapping

- **Sensor interpretation**
 - How do we **extract relevant information** from raw sensor data?
 - How do we represent and **integrate** this information **over space and time**?
- **Robot locations have to be estimated**
 - How can we identify that we are at a **previously visited place**?
 - How can we identify that we observe a **previously observed feature**?
 - This problem is the so-called **data association problem**.



Occupancy Grid Mapping

- Introduced by Moravec and Elfes in 1987-89
- ***Represent environment by a grid.***
- ***Estimate the probability that a location (a grid cell) is occupied by an obstacle.***
- **Key assumptions**
 - Occupancies of individual cells $m^{[xy]}$ are independent

$$\begin{aligned} Bel(m_t) &= P(m_t \mid z_{1:t}, x_{1:t}) \\ &= \prod_{x,y} Bel(m_t^{[xy]}) \end{aligned}$$

- Robot positions are known!



Occupancy Grid Mapping

- Problem becomes: Binary Estimation with Static State (Binary Bayes Filter)

```
1:  Algorithm occupancy_grid_mapping( $\{l_{t-1,i}\}, x_t, z_t$ ):  
2:      for all cells  $m_i$  do  
3:          if  $m_i$  in perceptual field of  $z_t$  then  
4:               $l_{t,i} = l_{t-1,i} + \text{inverse\_sensor\_model}(m_i, x_t, z_t) - l_0$   
5:          else  
6:               $l_{t,i} = l_{t-1,i}$   
7:          endif  
8:      endfor  
9:      return  $\{l_{t,i}\}$ 
```

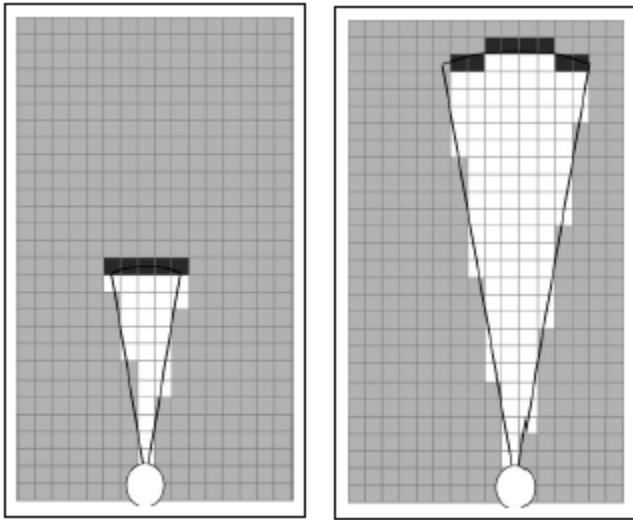
$$l_{t,i} = \log \frac{p(\mathbf{m}_i \mid z_{1:t}, x_{1:t})}{1 - p(\mathbf{m}_i \mid z_{1:t}, x_{1:t})}$$

$$l_0 = \log \frac{p(\mathbf{m}_i = 1)}{p(\mathbf{m}_i = 0)} = \log \frac{p(\mathbf{m}_i)}{1 - p(\mathbf{m}_i)}$$



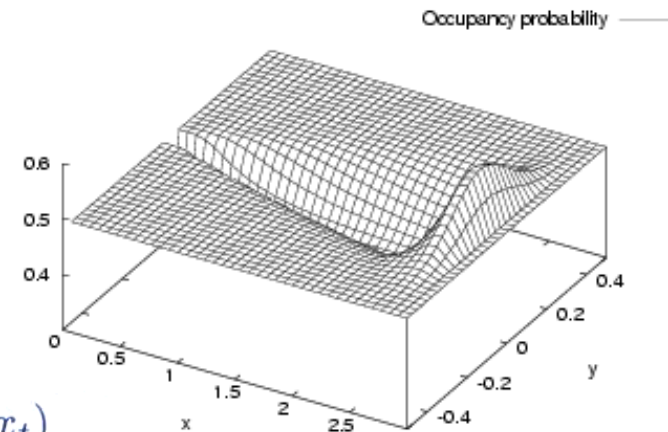
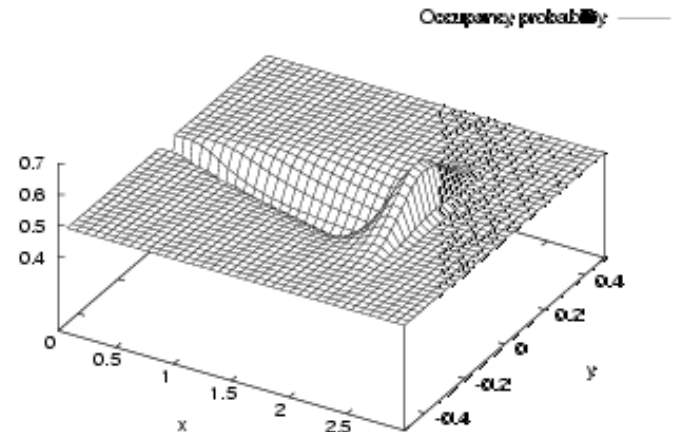
Inverse Measurement Model

- What is the occupancy likelihood of grid cells inside the measurement cone given that a set of “ranges” are measured



- Two examples given

$$\text{inverse_sensor_model}(\mathbf{m}_i, x_t, z_t) = \log \frac{p(\mathbf{m}_i \mid z_t, x_t)}{1 - p(\mathbf{m}_i \mid z_t, x_t)}$$





Inverse Measurement Model

```
1:   Algorithm inverse_range_sensor_model( $m_i, x_t, z_t$ ):
2:       Let  $x_i, y_i$  be the center-of-mass of  $m_i$ 
3:        $r = \sqrt{(x_i - x)^2 + (y_i - y)^2}$ 
4:        $\phi = \text{atan2}(y_i - y, x_i - x) - \theta$ 
5:        $k = \text{argmin}_j |\phi - \theta_{j,\text{sens}}|$ 
6:       if  $r > \min(z_{\text{max}}, z_t^k + \alpha/2)$  or  $|\phi - \theta_{k,\text{sens}}| > \beta/2$  then
7:           return  $l_0$ 
8:       if  $z_t^k < z_{\text{max}}$  and  $|r - z_{\text{max}}| < \alpha/2$ 
9:           return  $l_{\text{occ}}$ 
10:      if  $r \leq z_t^k$ 
11:          return  $l_{\text{free}}$ 
12:      endif
```

Thickness
of the
obstacles

Width of
the sensor
beam

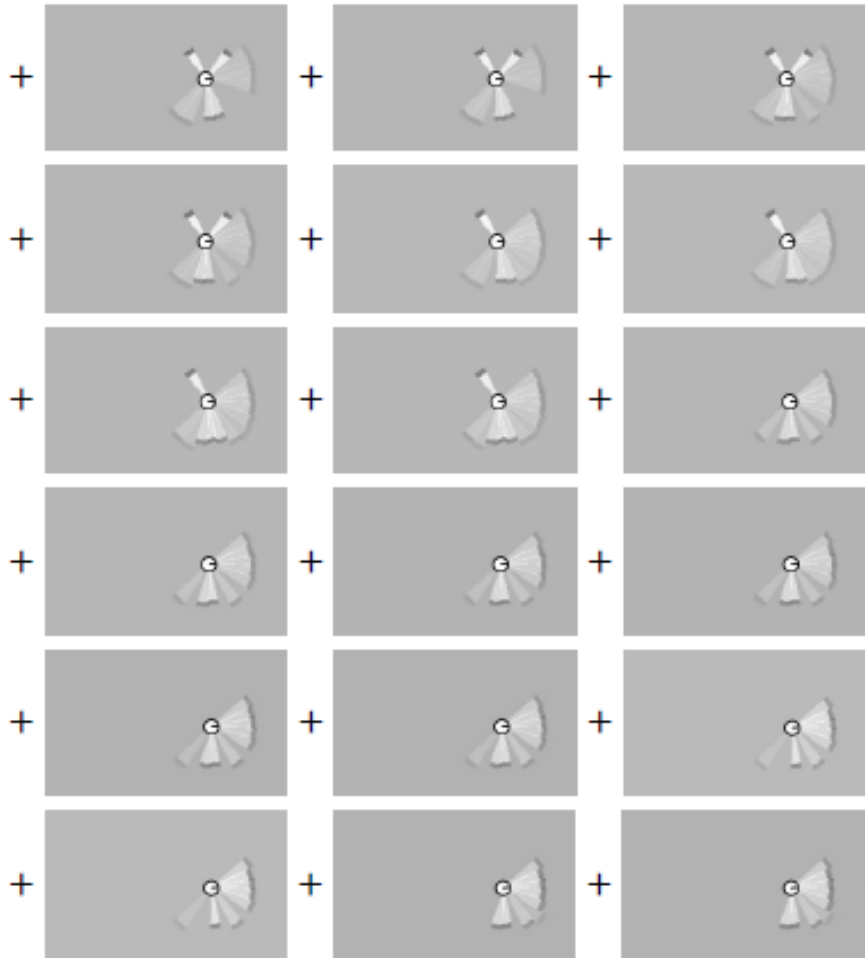
A simple inverse measurement (sensor) model.



Example



Initial Map



“Local Maps” from *Inverse Sensor Model*



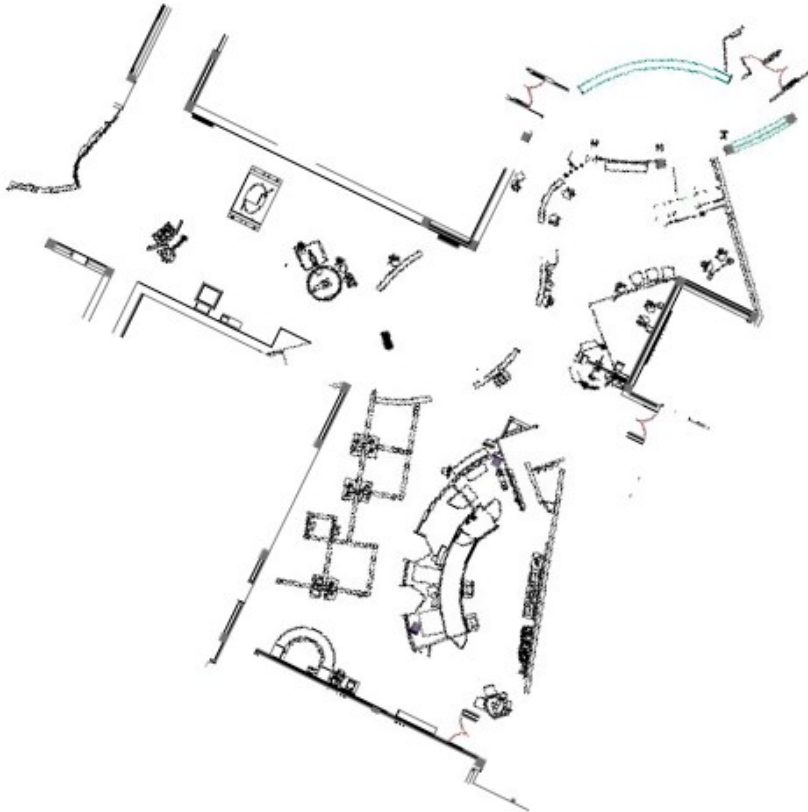
Completed *Occupancy Grid Map*
of office environment using sonar



Final Map



Blueprints vs Occupancy Maps

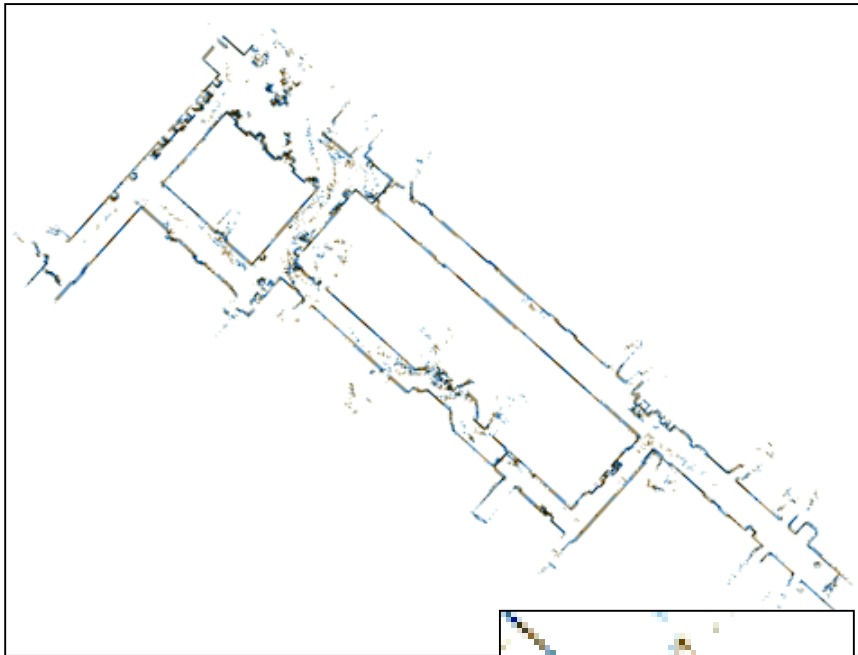


- Occupancy grid map shows “current state” of the environment as seen by the sensor of the robot,
- Clearly: There are differences with the “blueprint” of the environment.

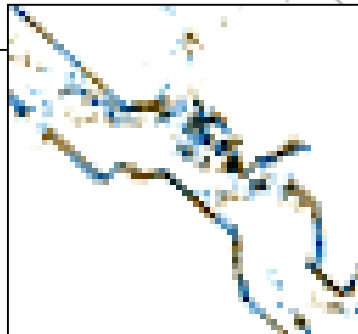


Raw Data vs Occupancy Maps

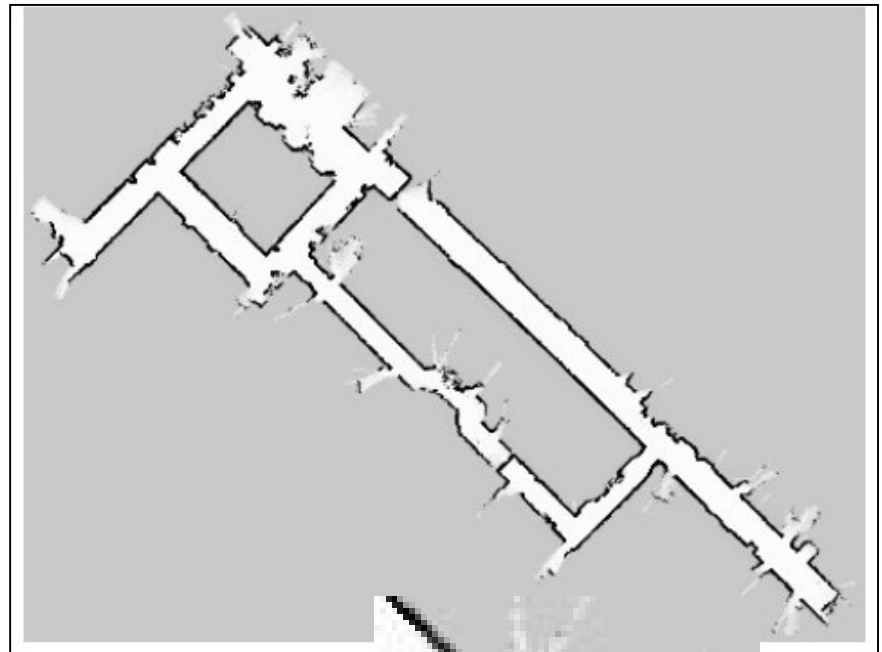
Raw laser range data with corrected pose



Laser scan data are isolated noisy “points in space”,
They can be corrupted by dynamic obstacles, people etc. in the environment.



Occupancy grid map



Occ. Grid map successfully filters out and “defined” the whole space.





“Robot body as a sensor”

- We may extend the inverse sensor model idea to the space occupied by the robot,
- If the robot is at a location x_t , that space and close vicinity must be “free”!!



Fusion of Multiple Sensors

- Information from multiple sensors can be transformed into “inverse sensor model”
- Camera (stereo-vision) example:

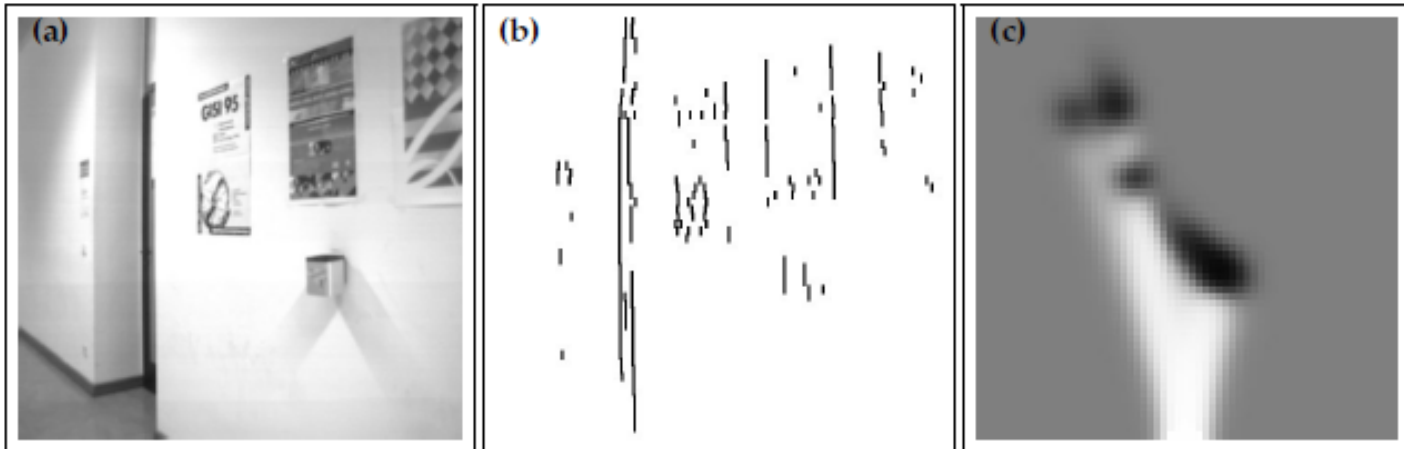


Image (pair)

Sparse disparity
(hence depth) map

Projection into 2D
map space and
smoothing



Fusion of Multiple Sensors

- Generally a good idea,
- Different sensor modalities may provide “independent” measurements,
- How to combine (fuse) them?
- If different sensors detect different obstacles, than simple integration with Bayes filter is ill-defined.
- In a naïve Bayesian approach: their update frequencies determines the result!!



Fusion of Multiple Sensors

- A popular approach: *Generate two separate maps and integrate them using a suitable combination function.*
- OR-Combination using De-Morgan Law:

$$p(\mathbf{m}_i) = 1 - \prod_k (1 - p(\mathbf{m}_i)^k)$$

$$a \vee b = \neg(\neg a \wedge \neg b)$$

- Compute Maximum (most pessimistic):

$$p(\mathbf{m}_i) = \max_k p(\mathbf{m}_i^k)$$



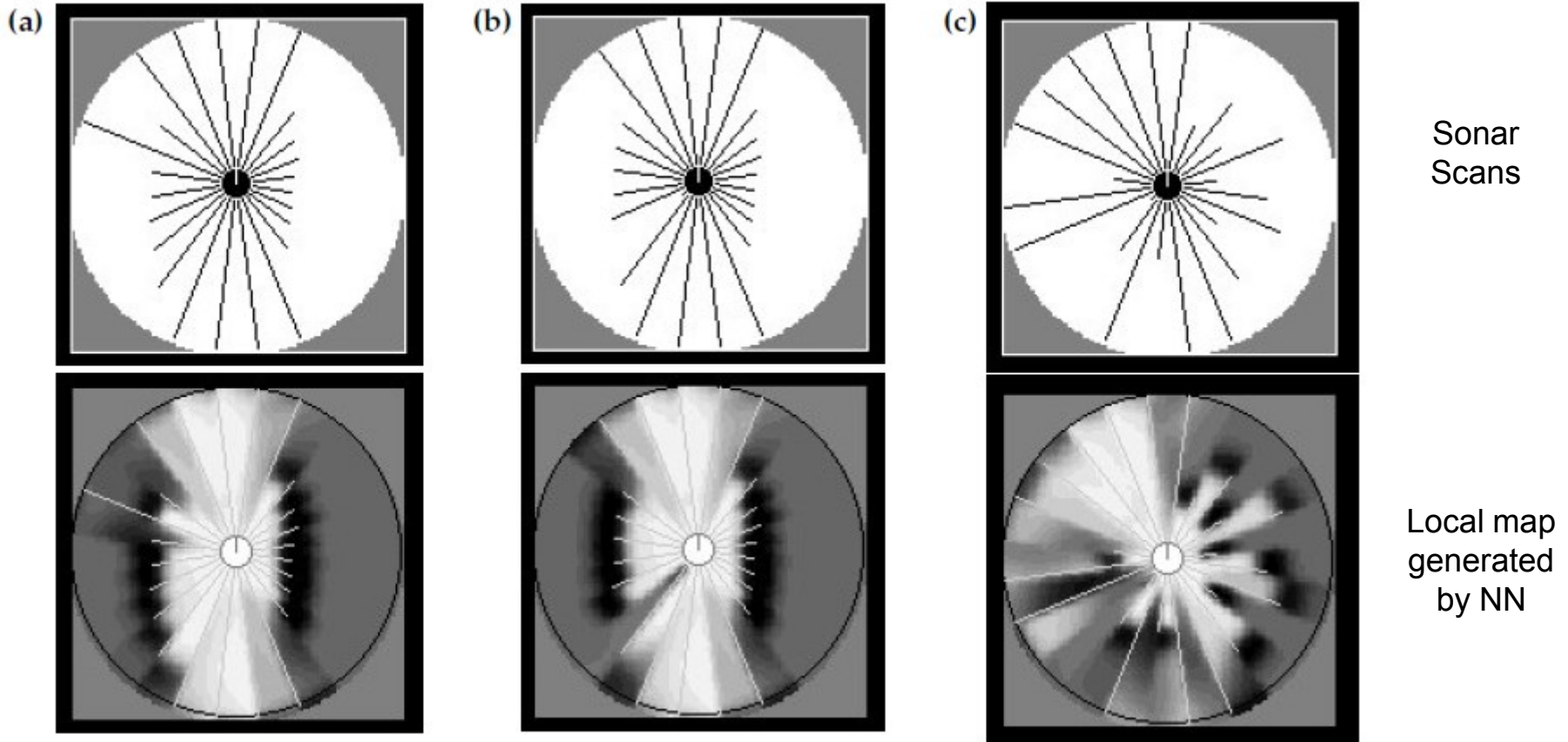
Learning the Inverse Sensor Model

- We may require a better sensor model than the “simple model” presented,
- One idea: Generate $\{x_t^{[k]}, z_t^{[k]}, m_i^{[k]}\}$ samples from either:
 - The forward measurement model,
 - Experiments with the actual sensor
- Then use a *nonlinear function approximator* such as a *Neural Network* with a *supervised learning algorithm* to learn the samples.
- You also need an *error function* for the supervised learning (e.g. *log likelihood of training data*).



Learning the Inverse Sensor Model

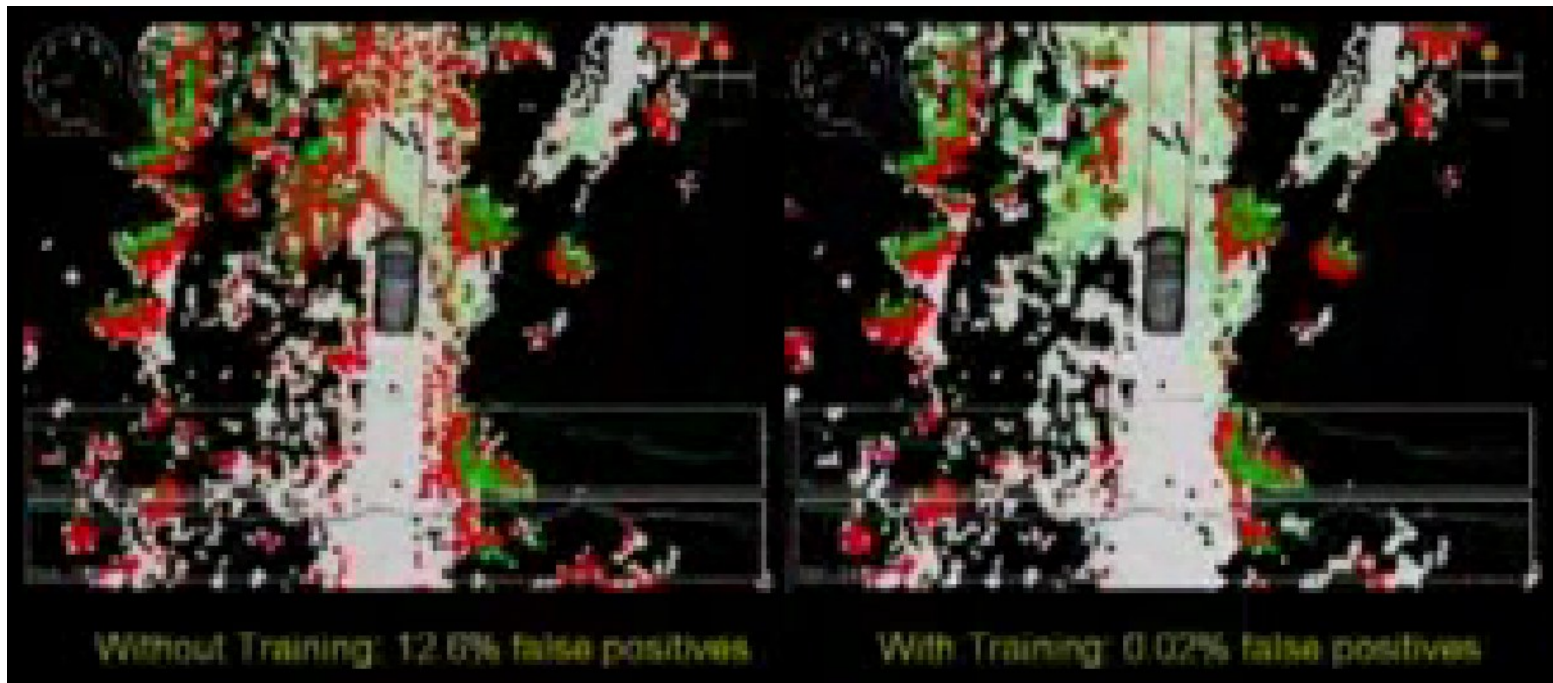
- An example





Learning the Inverse Sensor Model

- An example was also given in the Darpa Grand Challenge video! (Local map generated by a Neural Network from range scan data)





Cell Independence Assumption

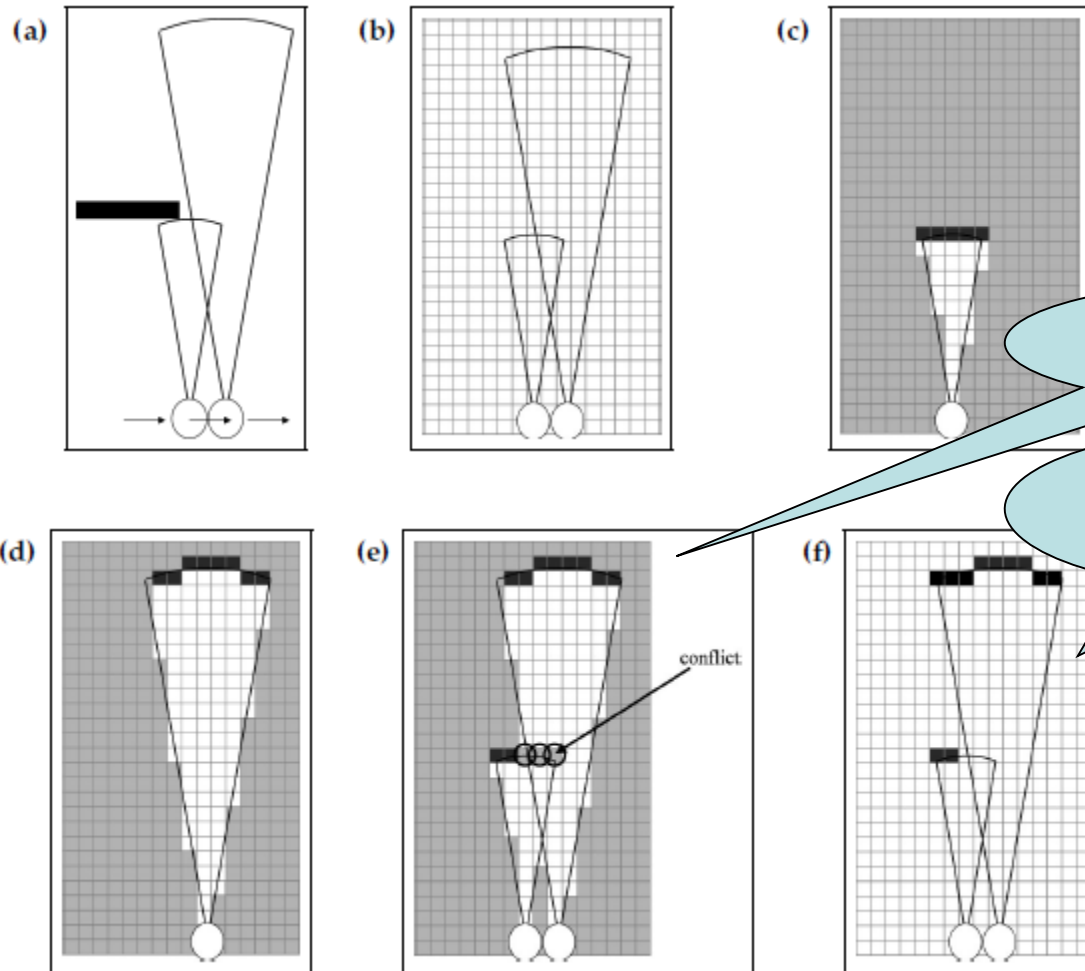
- To develop the occupancy grid mapping, we made the following independence assumption:

$$p(m \mid z_{1:t}, x_{1:t}) = \prod_i p(\mathbf{m}_i \mid z_{1:t}, x_{1:t})$$

- This is a strong assumption that may lead to conflicts in map generation
- Consider the following example:



Cell Independence Assumption



Resulting Occ. Grid Map

Another reasonable map that explains the measurement

The conflict arises from the Inter-Cell Independence assumption we have made!



Maximum A Posteriori Occ. Mapping

- An algorithm that uses the Forward Sensor Model,
- And computes the “Mode (peak) of the Posterior”:

$$m^* = \operatorname{argmax}_m \log p(m \mid z_{1:t}, x_{1:t})$$

- Which can be shown to be equivalent to:

$$m^* = \operatorname{argmax}_m \sum_t \log p(z_t \mid x_t, m) + l_0 \sum_i m_i$$

- This maximization can be done by an iterative hill-climbing technique



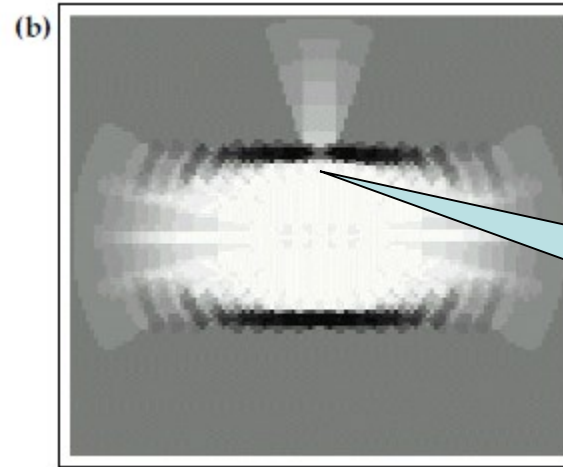
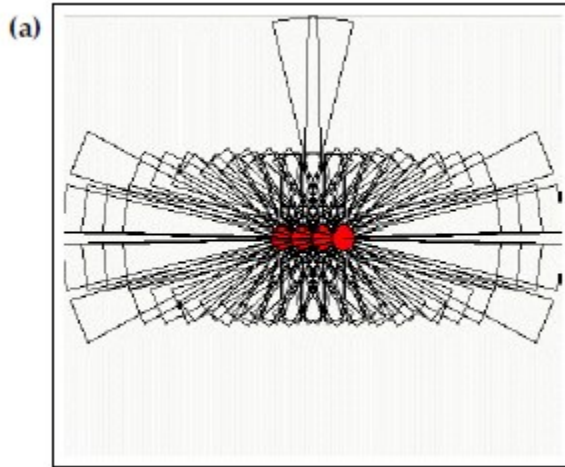
Maximum A Posteriori Occ. Mapping

```
1:  Algorithm MAP_occupancy_grid_mapping( $x_{1:t}, z_{1:t}$ ):  
2:      set  $m = \{0\}$   
3:      repeat until convergence  
4:          for all cells  $m_i$  do  
5:               $m_i = \operatorname{argmax}_{k=0,1} k l_0 + \sum_t \log$   
                                     measurement_model( $z_t, x_t, m$  with  $m_i = k$ )  
6:          endfor  
7:      endrepeat  
8:      return  $m$ 
```

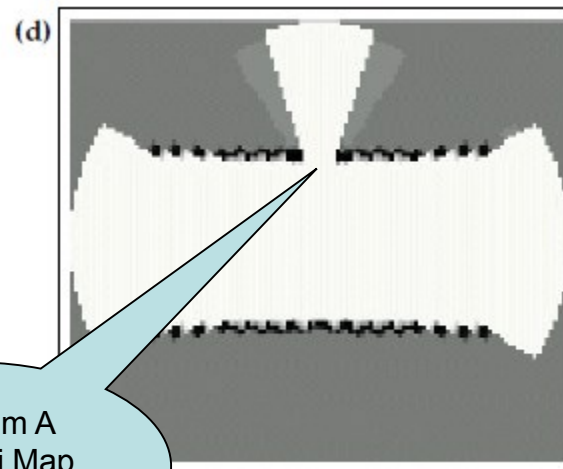
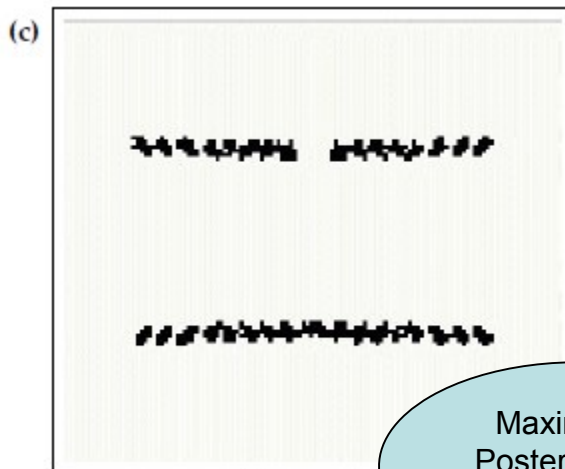
Note that the algorithm requires the entire measurement data set!!



Example: Maximum A Posteriori



Door have
disappeared
from Occ. Grid
Map



Maximum A
Posteriori Map
preserving the
Door



Summary

- We investigated mapping, assuming *robot pose is known exactly*,
- Standard occupancy grid mapping assumes independence between map cells and hence estimates posterior probability of occupancy,
- Data from multiple sensors can be combined either by using a single map updated by a Bayes filter (problematic) or by maintaining multiple maps (preferable),
- Multiple maps can then be combined probabilistically,
- Occupancy grid mapping relies on “Inverse Sensor Models” (from effects to causes) differently from Localization applications,



Summary (Continued)

- Inverse models can be learned from forward models by means of sampling and approximation OR by sampling the sensor (by a supervised learning based approximator)
- Independence assumption can be partially lifted (hence avoiding certain problems) by means of Maximizing the Posterior of the map.
- The posterior cannot be computed directly but still a maximizing map can be found through iterative optimization.
- This results in a map which is more consistent with the measurement data,
- Occupancy grid maps are very popular in robotics and have been successfully applied.